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The Zero-Trust Proctor: A Multi-Modal Architecture for Autonomous AI Interview Fraud Prevention

Architectural Framework: The Hybrid Edge-Cloud Model for Real-Time Performance

The foundational challenge in designing a fully autonomous, real-time AI proctoring system lies in reconciling the demand for millisecond-level responsiveness with the computational complexity of multi-modal AI analysis [63]. A purely cloud-centric architecture introduces network latency that is incompatible with the requirement for immediate in-session interventions, such as pausing an interview or blocking a connection [13]. This latency would not only delay critical actions but could also cause desynchronization between video streams and analytical data, undermining the entire system's credibility [145]. Consequently, the architectural philosophy must pivot to a Hybrid Edge-Cloud model, which strategically partitions responsibilities between the client device and the backend infrastructure to optimize for both speed and scale . This model is not merely an optimization choice but a non-negotiable prerequisite for achieving the project's core objective of zero-latency enforcement. The edge component acts as the frontline sensor and decision-maker, executing latency-sensitive tasks locally, while the cloud serves as the central brain for high-level aggregation, long-term storage, and complex analytics .

The client-side, or Edge, architecture is engineered for extreme efficiency and low latency. Its primary function is to process raw sensory data—video, audio, and user input—in real-time without relying on a constant, low-latency connection to a remote server. The specified technologies for this layer include WebAssembly (Wasm) and WebGPU, which allow for the deployment of highly optimized machine learning models directly within the browser environment [13] [62]. These frameworks enable near-native code execution, allowing computationally intensive tasks like computer vision inference to run efficiently on consumer-grade hardware [64]. The use of quantized ML models, delivered via TensorFlow.js or ONNX Runtime Web, further reduces the memory footprint and accelerates inference speed, making it feasible to run multiple simultaneous analyses on a single device [62]. For instance, lightweight object detection models like YOLOv26 are

specifically designed for edge-first deployment, offering predictable latency and stability crucial for real-time applications [12]. This on-device processing paradigm is a cornerstone of the system's design, as it keeps sensitive biometric data, such as facial landmarks and gaze vectors, confined to the candidate's machine until a security event necessitates its transmission [13]. This approach significantly mitigates privacy risks and reduces the bandwidth required for the application. The TRD correctly sets a target latency of less than 200 milliseconds for CV inference on the client side, a benchmark achievable with modern browsers and hardware acceleration via GPUs [62]. Research into object detectors has demonstrated latencies as low as 35 milliseconds per frame, validating the feasibility of this real-time constraint [11].

The responsibilities of the Edge AI engine are strictly limited to tasks requiring immediate feedback. This includes active liveness detection, where the system prompts the candidate to follow randomized screen trajectories to validate pupillary response and micro-movements, thereby defeating static photo and simple replay attacks [144]. Continuous passive eye tracking, using models like MPIIGaze, must also occur locally to monitor gaze vectors, fixation duration, and saccades in real-time [28]. Similarly, skeletal tracking for pose and gesture recognition (utilizing MediaPipe Pose or MoveNet) and initial object detection (e.g., for phones or secondary screens) are handled at the edge [25]. If the local processing load becomes too high—for example, if the frame rate drops below a threshold of 15 FPS—the system is designed to alert the user to close background applications, ensuring consistent performance . All telemetry generated by these edge processes, formatted as compact JSON payloads using MessagePack or Protobuf, is streamed to the cloud via a dedicated WebSocket endpoint (`/ws/v1/telemetry`) . This stream of consciousness from the candidate's device provides the raw material for the cloud's higher-level analysis [145].

The Cloud Backend, conversely, assumes the roles of data aggregation, complex analytics, archival storage, and providing the recruiter-facing interface. It is built on a scalable microservices architecture, typically deployed on a Kubernetes cluster that can auto-scale to handle variable loads . High-throughput data ingestion is managed by technologies like Apache Kafka, which is essential for processing the massive volume of time-series telemetry streaming from potentially thousands of concurrent interviews [63] [145]. Once ingested, specialized processing workers, likely written in Python with a FastAPI framework, execute more computationally demanding algorithms . These may include heavy NLP models for detailed audio transcription and analysis, or complex anomaly detection algorithms that correlate events across different modalities over the course of an entire session. The backend is also responsible for orchestrating the final stages of the Integrity Scoring Algorithm, which synthesizes the disparate telemetry streams into a

single, actionable score . For scalability and specialized computation, the backend provision includes GPU nodes (e.g., NVIDIA A10G), which can be reserved for fallback processing when client hardware is insufficient or for running intensive forensic analyses on recorded sessions [64] .

Data security and retention are paramount in the cloud architecture. Video recordings and raw biometric telemetry are stored in secure object storage, such as AWS S3, with strong encryption at rest using keys managed by a Key Management Service (KMS) . Access to these sensitive artifacts is governed by strict IAM policies and granted only through pre-signed URLs that expire after a short period, such as 15 minutes, to minimize exposure . Furthermore, the system must adhere to data retention policies mandated by regulations like GDPR and NIST SP 800-53, which often require audit records to be kept for a minimum of 90 days for incident investigation purposes [130][208]. To manage costs and privacy, automated processes, such as cron jobs or AWS Lambda functions, are implemented to purge raw video and telemetry data after the retention period expires, while preserving aggregated data points like the final integrity score and timestamped risk events for historical reporting and compliance audits [109]. The API gateway, which exposes endpoints like `/api/v1/session/init` and `/api/v1/integrity/report`, uses GraphQL for flexible data fetching on the frontend and REST for webhook integrations, ensuring a robust and adaptable interface for the application and any third-party systems . This hybrid architecture thus creates a resilient, scalable, and secure foundation that meets the stringent real-time demands of live proctoring while providing the centralized intelligence needed for comprehensive oversight and accountability.

Component	Technology/Standard	Role in Architecture	Rationale
Client-Side Inference	WebAssembly (Wasm), WebGPU, TensorFlow.js [13]	Runs lightweight ML models (Face Mesh, Gaze Estimation, Pose) locally for real-time analysis.	Achieves sub-500ms latency, keeps raw biometric data on-device, reducing privacy risks and bandwidth [13].
Real-Time Telemetry Streaming	WebSockets, Protobuf/ MessagePack	Transmits compressed `[timestamp, gaze_vector, head_pose]` data from the client to the cloud.	Provides a persistent, low-latency channel for continuous data ingestion, essential for live monitoring [63].
Backend Ingestion	Apache Kafka	Handles high-throughput, time-series telemetry streams from thousands of concurrent sessions.	Ensures reliable and scalable buffering of real-time data before it is processed by downstream services [63] [145].
Database - Time-Series	TimescaleDB / InfluxDB	Stores high-frequency biometric data (eye gaze, head pose) optimized for fast writes and time-based queries.	Designed for time-series data, offering superior performance for logging telemetry compared to standard relational databases .
Database - Relational	PostgreSQL	Manages structured data like user profiles, session metadata, and risk event logs.	Provides ACID compliance and robust querying capabilities for transactional and reference data .
Data Storage	AWS S3 (encrypted)	Securely stores final video recordings and generated PDF reports.	Offers durable, scalable, and secure object storage with fine-grained access control and KMS-managed encryption .
Security Standard	ISO 27001, SOC 2 Type II, GDPR, CCPA	Governs data classification, access controls, retention, and protection across the entire system.	Ensures compliance with international information security and privacy regulations, building trust and mitigating legal risk [106] [129] [208].

Identity & Anti-Spoofing Engine: Defeating Impersonation and Synthetic Media

The most fundamental threat to the integrity of an AI-powered interview is impersonation, where an unqualified individual substitutes for the legitimate candidate. The proctoring system must therefore feature an exceptionally robust Identity & Anti-Spoofing Engine capable of distinguishing a live, bona fide candidate from a wide array of presentation attacks, including printed photos, video replays, 2D/3D masks, and increasingly sophisticated deepfakes [27] [56]. Achieving the goal of a >98% detection rate for spoofing requires a multi-layered defense strategy grounded in established biometric standards and cutting-edge research in synthetic media forensics [4]. This engine operates continuously throughout the session, verifying identity passively while also conducting periodic active checks to thwart injection attacks.

A critical first line of defense is **Active Liveness Detection**, which involves prompting the candidate to perform random, unpredictable actions. The proposed method of displaying a moving dot on a randomized trajectory and asking the candidate to follow it with their eyes is a powerful technique . This forces the candidate to engage in dynamic micro-movements and induces a natural pupillary light reflex—the change in pupil diameter in response to shifts in screen brightness—which is impossible to replicate with a static image or a simple mask [56] [144]. This form of active verification aligns with the principles of Presentation Attack Detection (PAD), which focuses on defending biometric systems at the point of capture (i.e., the camera sensor) [53] . The industry benchmark for evaluating the efficacy of PAD systems is the **ISO/IEC 30107 series of standards** [47] [139]. Compliance with this standard, particularly Level 2 or Level 3, is considered "table stakes" for serious proctoring solutions, as it ensures the system has been rigorously tested against a diverse set of spoofing attacks under controlled conditions [140]. These tests yield two key metrics: the Attack Presentation Classification Error Rate (APCER), which measures the failure to detect an attack, and the Bona Fide Presentation Classification Error Rate (BPCER), which measures false rejection of a genuine user [141] [176]. Leading systems aim for extremely low APCER values, often below 1.0%, demonstrating a high degree of resistance to known spoofing methods [175]. Certifications from accredited labs like iBeta serve as a strong indicator of a system's anti-spoofing capability; for example, Shufti achieved a 0% APCER and 0% BPCER in Level 3 passive liveness testing, while Facia scored a perfect 0% APCER in Level 2 PAD testing [51] [52] .

While active checks provide strong verification at key moments, continuous passive liveness detection is essential for ongoing session integrity. This involves analyzing subtle physiological cues in the live video feed. One of the most promising techniques is the analysis of remote photoplethysmography (rPPG), which estimates a person's pulse by detecting minute changes in skin color caused by blood flow in the face [117]. A deepfake video, even if visually convincing, will almost certainly fail to replicate these real-time physiological signals, providing a definitive way to identify synthetic media [54] . Another powerful passive technique is the detection of temporal inconsistencies between audio and visual signals, a common flaw in lip-sync deepfakes. Research has produced methods like LIPINC (LIP-syncing detection based on mouth INConsistency) that exploit these mismatches, achieving detection accuracies of over 95% [114] [115]. By comparing the timing of lip movements with the corresponding audio waveform, the system can spot fakes where the mouth shapes are inconsistent with the underlying speech [112] [132]. This multi-modal approach, integrating audio and video analysis, dramatically increases the difficulty for attackers, as it requires them to manipulate multiple signal streams simultaneously and coherently [117] [131].

The anti-spoofing arsenal must also extend to audio. Deepfake voice technology has advanced rapidly, making audio-based identity verification alone insufficient. The system must employ dedicated audio forensics to detect synthetic speech and audio splicing. This can involve analyzing acoustic features for anomalies, such as unnatural prosody or spectral characteristics that differ from human speech [185][187]. Recent research has shown that combining powerful speech recognition models like OpenAI's Whisper with specialized architectures like AASIST (a model designed for audio antispoofing) can enhance the detection of deepfake audio [124]. Furthermore, the inherent properties of speech itself can be leveraged for forgery detection; genuine speech tends to have a smooth, coherent structure, whereas forged audio may exhibit subtle discontinuities that are imperceptible to humans but detectable by a well-trained model [162]. Speech foundation models, though not explicitly trained for forgery detection, can implicitly encode a forensic signal related to the authenticity of the speaker's vocal tract [162]. By integrating these audio analysis techniques with the visual liveness checks, the system creates a formidable barrier against identity fraud. The table below summarizes the spoofing attack types and the corresponding countermeasures the engine should deploy.

Spoofing Attack Type	Description	Primary Countermeasure(s)	Supporting Technologies/Standards
Photo Replay	Using a printed photograph of the candidate's face.	Active Liveness (pupillary light reflex, micro-movements). Passive Analysis (lack of physiological signals).	ISO/IEC 30107-3 [47], Pupillary Response Analysis [144], rPPG for heartbeat detection [117].
Video Replay	Playing a pre-recorded video of the candidate.	Active Liveness (randomized motion prompts). Audio-Visual Sync Analysis.	Moiré pattern detection , Audio-Visual Drift Analysis [60], Randomized Trajectory Tracking .
2D/3D Masks	Using a plaster, silicone, or other physical replica of a face.	Active Liveness (micro-expressions, texture analysis). Passive Analysis (lack of physiological signals).	ISO/IEC 30107-3 [47], Micro-expression Analysis [54], Periocular Area Analysis [50].
Lip-Sync Deepfakes	Synthesizing a video of a person speaking words they never said, using another source.	Temporal Audio-Visual Mismatch Analysis.	LIPINC Framework [115], Lip-Forgery Detection [112], Multi-modal Consistency Checks [131].
Synthetic Voice	Generating audio using text-to-speech or voice cloning technology.	Acoustic Anomaly Detection. Speech Coherence Analysis.	Whisper + AASIST Fusion [124], Segmental Speech Feature Analysis [199], Prosody and Spectral Analysis [185].
Deepfake Video (General)	Advanced end-to-end synthesized video of a person.	Multi-modal Consistency Check. Physiological Signal Analysis (rPPG).	rPPG for Heartbeat [117], Multi-sensor fusion (camera, LiDAR, IMU) [1], Deep Learning Forensic Models [210].

Behavioral Biometrics Suite: Detecting Cognitive Load and Anomalous Actions

Beyond verifying identity, the proctoring system must act as a vigilant observer of the candidate's behavior, as actions are often more revealing than appearances. A sophisticated suite of behavioral biometrics is essential for inferring cognitive state, detecting attempts at cheating, and flagging anomalous actions in real-time. This suite integrates continuous passive monitoring of eye movements, head pose, and hand gestures with advanced audio analysis to build a holistic profile of the candidate's engagement and potential misconduct. The goal is to move beyond simple presence detection to a nuanced understanding of behavior, identifying patterns indicative of stress, distraction, or collaboration with external agents. The system's ability to distinguish between normal nervous habits and deliberate cheating maneuvers hinges on the accuracy and contextual awareness of these behavioral models.

Eye tracking is arguably the most powerful tool in the behavioral suite, as it provides direct insight into a candidate's focus and thought processes [29]. The system must employ continuous passive eye tracking to monitor several key metrics: gaze vector, fixation duration, saccades (rapid eye movements), and pupil dilation [28]. Prolonged off-screen gazes are a classic indicator of cheating, suggesting the candidate is consulting unauthorized materials like hidden notes, a secondary screen, or a phone [61] [72]. The system should be configured to raise a low-level flag if gaze deviates from the designated area for more than three seconds, triggering a gentle UI prompt before logging a formal risk event . Conversely, erratic saccades and prolonged fixation on a single point off-screen can indicate intense concentration on a hidden prompt or script [28]. Head pose estimation, using models like MediaPipe Pose, complements eye tracking by monitoring the orientation of the candidate's head in three dimensions (yaw, pitch, roll) [18]. A sudden turn of the head away from the primary camera feed could signify interaction with a second person or a hidden device. Some studies have explored simple, efficient algorithms for head pose detection, focusing on horizontal and vertical orientation to estimate glances away from the screen [21]. The combination of gaze and head pose provides a much richer picture of the candidate's attentional focus than either modality alone.

Hand gesture and pose recognition add another layer of behavioral analysis, enabling the system to detect interactions with the physical environment. Using skeletal tracking algorithms like MediaPipe Pose or OpenPose, the system can map the candidate's body and limbs in real-time [25] [136]. This allows it to detect several high-risk behaviors. For

instance, it can identify if the candidate's hands leave the visible frame, a potential sign of manipulating a hidden object. More specifically, it can recognize hand-to-ear gestures, which are a strong indicator of an attempt to use a hidden wireless earpiece [119]. The system can also be trained to recognize the manipulation of unauthorized objects, such as a smartphone that might appear in the peripheral view or be reflected in a monitor's surface [120]. The accuracy of gesture recognition is critical; one study reported an overall classification accuracy of 83.06% using a compact gesture system, demonstrating the feasibility of this approach [25]. By tracking the full body, the system can also detect the presence of a secondary person in the room, who might be providing answers or instructions to the candidate [27].

Audio analysis forms the final pillar of the behavioral suite, acting as a sensitive microphone for the sounds of cheating. The system must perform continuous background noise analysis to detect a range of suspicious audio cues. The sound of keyboard typing that is not synchronized with the candidate's speech cadence is a strong indicator of ghostwriting or receiving real-time assistance [40, 72]. This technique falls under the category of Acoustic Side-Channel Attacks (ASCAs), where the subtle sounds of keystrokes are analyzed to reconstruct typed content [45, 74]. Remarkably, these attacks can be carried out using just a standard smartphone microphone placed in the vicinity, making them a credible threat in an unsupervised setting [99, 201]. Advanced deep learning models can achieve very high accuracy in classifying keystrokes from ambient noise, even in noisy environments [100, 184]. Beyond typing, audio proctoring listens for secondary voices, such as whispers, conversations, or reading aloud from a script, all of which are clear signs of unauthorized assistance [70, 157]. Modern Automatic Speech Recognition (ASR) systems like OpenAI's Whisper, especially newer versions like WhisperX that provide word-level timestamps, can be used to transcribe audio and potentially identify the presence of a second speaker whose language or accent differs from the candidate's [81, 83]. By fusing these multimodal behavioral signals—where a glance off-screen is correlated with the sound of keyboard typing and a hand gesture toward the ear—the system can build a compelling case for cheating with a high degree of confidence, enabling the strict enforcement protocols required by the research goal.

Environmental Scanning Module: Securing the Candidate's Physical Space

To create a truly secure interview environment, the proctoring system must extend its vigilance beyond the candidate's face and behavior to encompass the entire physical space captured by the webcam. Unauthorized materials, secondary devices, and covert communication tools represent significant cheating vectors that must be actively scanned for and detected in real-time. The Environmental Scanning Module is therefore a critical component of the system, employing a combination of computer vision, audio analysis, and advanced sensing technologies to build a complete picture of the candidate's surroundings. While standard object detection can identify obvious threats, the module must also incorporate more sophisticated techniques like RF fingerprinting and thermal imaging to counter advanced, covert cheating methods. This multi-pronged approach ensures that the system can effectively neutralize attempts to gain an unfair advantage through environmental manipulation.

The most direct method of environmental scanning is through computer vision-based object detection. Using a lightweight, edge-deployable model like YOLOv26, the system can analyze the video feed in real-time to identify and classify objects within the candidate's field of view [12] [120]. The primary targets for detection would include prohibited items such as mobile phones, tablets, secondary computer monitors, books, notebooks, or any other unauthorized materials [134]. The system should not only detect the presence of these objects but also track their appearance and disappearance over time, flagging instances where a phone is briefly obscured or moved. Advanced implementations can analyze reflections in windows or glossy surfaces to detect secondary screens that might not be directly visible to the main camera [120]. The integration of a 360-degree environmental scan during the onboarding phase, where the candidate pans their camera around the room, provides a baseline of the legitimate environment and helps the system differentiate between authorized and unauthorized objects . This proactive scanning prevents candidates from introducing cheating aids mid-interview.

Audio analysis serves as a powerful complementary tool for environmental scanning. As previously discussed, it can detect the sound of keyboard typing associated with ghostwriting [45] [74]. However, its utility extends to detecting the presence of electronic devices through their unique acoustic signatures. Furthermore, advanced audio forensics can be used to detect secondary persons in the room, whose voices may be too quiet for the candidate to hear but are easily picked up by the sensitive microphone [157]. The system should be able to distinguish between the candidate's voice and background noise,

isolating and analyzing any other speech-like activity. The development of outlier-robust multivariate time series models for audio anomaly detection enables the system to identify previously unseen anomalous sounds, such as the hum of a hidden recording device or the click of a smartwatch [73]. By correlating these audio findings with the visual feed, the system can confirm the location and nature of the detected sound source.

To stay ahead of sophisticated cheaters, the Environmental Scanning Module must incorporate advanced sensing technologies that go beyond standard RGB cameras and microphones. **Thermal Imaging** is a particularly potent countermeasure. Hidden electronic devices, even when turned off, retain a residual heat signature. Thermal cameras can detect these "thermal ghosts," revealing the precise location of recently used phones, laptops, or surveillance equipment [6] [7]. This makes it nearly impossible to hide powered-down electronics in the room. The system could integrate a thermal camera as a premium feature or leverage dual-camera setups to enable real-time depth estimation and thermal analysis, further enhancing its environmental awareness [122][123]. Another advanced technique is **Radio Frequency (RF) Fingerprinting**. Every electronic device emits a unique RF signature due to minor manufacturing variations in its components [153]. Systems like RFScan can passively detect, fingerprint, and localize concealed IoT devices by analyzing these signatures in the indoor environment [92] [121]. This technology can identify hidden wireless earpieces or smartwatches, even if they are in airplane mode, by recognizing their unique hardware fingerprint on the ambient RF spectrum [94] [188]. Finally, **WiFi-based sensing** offers another layer of environmental awareness. WiFi signals are affected by the presence and movement of people. Systems like WiPIHT can use commercial WiFi devices to track human activity trajectories, allowing the system to detect if a second person enters the room or moves around, even without a direct line of sight [118][154]. By integrating these advanced sensors, the system creates a multi-layered, defense-in-depth strategy that transforms the candidate's physical environment from a potential source of cheating into a verifiable and secure space.

Sensing Method	Principle of Operation	Detected Threats	Strengths & Limitations
RGB Object Detection	Analyzes visual pixels to identify and classify objects in the video feed.	Phones, secondary monitors, books, unauthorized notes.	Strong for obvious objects; can be fooled by occlusion or reflective surfaces. Models like YOLOv26 are suitable for edge deployment [12] [120].
Acoustic Analysis	Listens for and analyzes sound waves from the microphone.	Keyboard typing (ghostwriting), secondary voices, whispers, electronic hums.	Highly effective for detecting ASCAs and unauthorized conversations; can be affected by ambient noise [45] [73] [157].
Thermal Imaging	Detects infrared radiation (heat) emitted by objects.	Recently used electronic devices (phones, laptops), hidden cameras, or people in dark rooms.	Excellent for finding powered-down electronics; requires specialized hardware [6] [7] [169].
RF Fingerprinting	Passively detects and analyzes unique radio frequency signatures of electronic devices.	Hidden wireless earpieces, smartwatches, IoT devices in airplane mode.	Can identify devices even when not actively transmitting data; requires an RF scanner [92] [94] [121].
WiFi-Based Tracking	Analyzes how WiFi signals are reflected and absorbed by people and objects.	Presence and movement of secondary individuals in the room.	Non-intrusive and works through walls; provides coarse positional data rather than a visual image [118] [154].

Enforcement Protocol: A Deterministic Rule-Based Intervention System

The user's directive for a "fully autonomous" system with "extremely strict enforcement protocols" necessitates a robust intervention mechanism that goes beyond the probabilistic outputs of AI models. While machine learning is essential for detection, the logic governing actions—warnings, pauses, and terminations—must be explicit, deterministic, and auditable. The single most common failure in production AI projects is deploying an agentic AI where a deterministic system was required; therefore, the enforcement protocol should be architected as a Complex Event Processing (CEP) engine integrated with a business rules engine, such as Drools [16] [179] [180]. This hybrid approach allows the system to react intelligently to real-time streams of data by applying a predefined set of rules that escalate in severity based on the type and confidence of detected events. This ensures that every action taken by the system is transparent, repeatable, and aligned with the zero-tolerance policy for integrity breaches.

The core of the enforcement protocol is a tiered system of alerts and interventions, designed to give the candidate opportunities to correct their behavior before facing severe consequences. This escalates from low-level flags to critical blocks, creating a graduated response to cheating attempts. The system must be designed to minimize false

positives, as an overly aggressive warning system can damage the candidate experience and erode trust [37] [43]. The implementation plan rightly identifies this as a key risk, proposing safeguards like grace periods and contextual AI to avoid unfairly penalizing legitimate behavior . The CEP engine processes the continuous stream of telemetry from the edge, looking for patterns and combinations of events that exceed predefined thresholds. When a potential infraction is detected, it is classified by type and severity.

The following table outlines a sample escalation protocol, mapping specific detected events to corresponding system responses. This protocol is designed to be configurable and auditable, forming the basis of the system's autonomous enforcement capabilities.

Severity Level	Triggering Event(s)	System Action	Rationale
Low	Gaze vector indicates looking off-screen for >3 seconds [61]. Head leaves the designated frame.	A non-punitive UI overlay appears, gently prompting the candidate to refocus (e.g., "Please keep your face within the frame") . No log entry is made at this stage.	Allows the candidate to self-correct without feeling punished. Avoids premature flagging for brief, accidental glances.
Medium	Confirmed secondary voice detected via audio analysis [157]. Hand-to-ear gesture recognized . Multiple low-level flags accumulate within a short timeframe.	A more formal warning is displayed on-screen. Audio may be muted for a short period (e.g., 15 seconds) to disrupt communication. A "low" risk event is logged with a timestamp.	Escalates the response for confirmed or repeated infractions. Disrupts the cheating attempt without immediately terminating the session.
High	Secondary monitor or phone is detected in the environment [120]. Keyboard typing is detected that doesn't match speech cadence [72].	The interview is autonomously paused. A message is displayed informing the candidate that a potential integrity breach has been detected and the session is on hold for review. A "medium" or "high" risk event is logged.	Halts the assessment immediately to prevent further potential cheating. Gives the system administrator or recruiter time to review the evidence before proceeding.
Critical	Confirmed deepfake or synthetic media attack is detected [67]. Acoustic Side-Channel Attack (keyboard typing) is conclusively proven [45]. A critical risk event is confirmed by the rules engine.	The interview is terminated. The session results are automatically locked and flagged as invalid. A "critical" risk event is logged, and an immediate alert is sent to the recruiter/administrator.	Enforces a zero-tolerance policy for the most severe and difficult-to-detect forms of cheating. Prevents fraudulent outcomes from being recorded or used in hiring decisions.

This rule-based system provides the necessary clarity and predictability for an enforcement protocol. Each action is tied to a specific, measurable condition, and the progression from warning to termination is logical and consistent. The use of a CEP engine is ideal for this workflow, as it is designed to process streams of events in real-time and trigger actions based on complex, time-based conditions [182]. For example, the rules engine could be configured to look for the conjunction of "secondary voice detected" AND "candidate's head is turned away from the camera" within a 5-second window, which would trigger a "high" or "critical" level response. This allows the system to understand context, rather than reacting to isolated events.

Furthermore, the system must be designed with contingency plans for technical failures. Client hardware limitations are a significant risk; if a candidate's device cannot support the edge AI workload, the system must have a fallback mode . This could involve relaxing some of the stricter monitoring (e.g., disabling high-resolution eye tracking) or shifting the processing load to the cloud, albeit with increased latency. Another critical contingency is the potential for privacy backlash or legal challenges [4] . To mitigate this, the system's architecture should prioritize privacy by design, keeping raw biometric data on-device and only transmitting it upon a confirmed critical event, thereby limiting the scope of data collection to what is absolutely necessary for enforcement [13] . By combining the pattern-recognition power of AI with the transparent, auditable logic of a deterministic rule engine, the system can deliver on the promise of fully autonomous, real-time, and strictly enforced proctoring.

Post-Session Forensics and Data Governance: Auditable Integrity and Compliance

A truly comprehensive proctoring system does not end when the interview concludes; its work is validated and solidified in the post-session forensic analysis phase. This module is responsible for transforming the raw, real-time telemetry stream into an auditable, immutable record of the session's integrity. It serves as the ultimate arbiter, allowing recruiters to validate the system's real-time decisions, uncover subtle incidents that may have been missed, and generate detailed reports for compliance and accountability purposes. The design of this module is inextricably linked to the overarching principles of data governance, which mandate strict adherence to privacy regulations like GDPR, BIPA, and CCPA, as well as security standards like ISO 27001 and NIST SP 800-53 [10][6][12][9]. A failure in data governance can render the system legally untenable and ethically indefensible, regardless of its technical prowess.

The core of the forensic module is the Recruiter Dashboard, which functions as a command center for reviewing flagged sessions. Its primary feature is a scrub-able video timeline overlaid with visual indicators of all detected risk events . When a recruiter hovers over a red marker on the timeline, the video player jumps to that exact moment and pauses, providing instant context. A side panel then displays the precise telemetry data that triggered the flag—for example, showing the exact gaze vector coordinates that indicated an off-screen look, the duration of the glance, and the corresponding audio spectrogram that may have shown keyboard noise . To further bridge the gap between

the AI's perception and the human reviewer's understanding, the dashboard should include a "Ghost View" toggle. This feature draws the MediaPipe skeletal wireframe and gaze vectors directly onto the video feed, allowing the recruiter to see exactly what the computer vision engine saw at any given moment . This transparency is crucial for building trust and ensuring that automated decisions are fair and accurate. The final output of this process is the Integrity Score—a single, 0-100 numerical value that summarizes the session's integrity, derived from the aggregation of all risk events and their severities [57] . This score distills a complex dataset into an intuitive metric for quick decision-making.

Robust data governance is the bedrock upon which this forensic capability is built. The system must be architected to comply with a complex web of global regulations concerning personal and biometric data. Under GDPR, data processing must have a lawful basis, and candidates must be informed about what data is collected and why [127] [129]. The Illinois Biometric Information Privacy Act (BIPA) imposes particularly strict requirements on the collection and use of biometric identifiers like face geometry . The system's database schema reflects these concerns: raw facial embedding vectors are stored in a vector database, but the relational database (PostgreSQL) only holds a cryptographic hash of the vector, preventing reverse-engineering . This principle of data minimization and hashing applies to all sensitive Personally Identifiable Information (PII).

Secure and compliant data retention is another critical aspect of governance. The system must implement automated data lifecycle policies. Raw video recordings and high-frequency biometric telemetry, while valuable for forensics, contain sensitive information that should not be stored indefinitely. The system should be configured to automatically delete these raw artifacts after a set retention period, such as 90 days, in line with recommendations from standards like NIST SP 800-53 [130]. During this retention period, all data must be encrypted at rest (e.g., using AWS S3 Server-Side Encryption with KMS keys) and in transit (via TLS 1.3) . Furthermore, all system activities—including session starts and ends, risk events, and administrative actions—must be logged securely. These audit trails are essential for accountability and are a requirement of standards like NIST SP 800-53 AU-2 and AU-9, which call for logging security-relevant events and ensuring audit logs are tamperproof and cryptographically verifiable [159] [160] [214]. By implementing immutable audit logs, the system creates an unalterable record that can withstand scrutiny and support investigations into any alleged misuse or error. Finally, to address the inherent privacy risks of behavioral data, the system should explore advanced privacy-preserving techniques. For gaze data, which can be re-identifying, methods like differential privacy can be applied to the anonymized telemetry before it is stored or used

for training new models, reducing the risk of identification [164][167][168]. By weaving these principles of transparency, security, and compliance into the fabric of the post-session analysis and data governance modules, the Aegis Interview Room establishes itself not only as a powerful anti-cheating tool but also as a responsible custodian of sensitive candidate data.

Reference

1. Camera, LiDAR, and IMU Based Multi-Sensor Fusion SLAM: A Survey https://www.sciopen.com/article/10.26599/TST.2023.9010010
2. [PDF] Ensuring-academic-integrity-through-automated-online-exam ... https://www.researchgate.net/publication/400765099_Ensuring_academic_integrity_through_automated_online_exam_proctoring_a_decade_long_systematic_review/fulltext/698f71ba7247bc6473e077b2/Ensuring-academic-integrity-through-automated-online-exam-proctoring-a-decade-long-systematic-review.pdf?origin=scientificContributions
3. [PDF] E- Proctoring in Theory and Practice : A Review https://www.qqi.ie/sites/default/files/2021-12/e-proctoring-in-theory-and-practice-a-review.pdf
4. [PDF] AI Proctoring: Academic Integrity vs. Student Rights - UC Law Journal https://www.uclawjournal.org/wp-content/uploads/10-Mita_final.pdf
5. Assistant for the detection of potential cheating behavior in ... https://educationaldatamining.org/EDM2025/proceedings/2025.EDM.short-papers.20/index.html
6. Can Thermal Imaging Cameras Detect Security Devices? What You ... https://www.pixfra.com/can-thermal-imaging-cameras-detect-security-devices-what-you-need-to-know
7. Choosing the Right Thermal Camera for Counter Surveillance - iSecus https://www.isecus.com/thermal-camera-for-counter-surveillance/
8. Deep Proctoring for Hidden Device Detection: Secure Exams. https://digiproctor.com/deep-proctoring
9. How to hide from a thermal imager: effective strategies and methods https://stg-defence.com/en/how-to-hide-from-a-thermal-imager-effective-strategies-and-methods/

10. Protect your privacy with Thermal Imaging pt 3 - SENSE Software https://sensesoft.eu/article.php?id=77&lang=en&page=0
11. [PDF] Real-Time AI-Driven Exam Cheating Detection Using YOLO and ... https://www.irejournals.com/formatedpaper/1712742.pdf
12. YOLOv26: An Object Detector Built for Real-Time Deployment https://learnopencv.com/yolov26-real-time-deployment/
13. AI Video Analytics for Online Learning in 2026 - Fora Soft https://www.forasoft.com/blog/article/ai-video-analytics-online-learning
14. Recruitix - Project Case Study - Debojyoti De Majumder https://www.djyo.tech/projects/recruitx
15. Introducing τ-voice Benchmark for Real-Time Voice Agents - LinkedIn https://www.linkedin.com/posts/brettaylor_%F0%9D%9C%8F-voice-benchmarking-real-time-voice-agents-activity-7456078030373949440-08e3
16. 7 Reasons AI Agents Fail in Production in 2026 - Brain Station 23 https://brainstation-23.com/why-most-ai-agents-fail-in-production/
17. A comprehensive review of the changing landscape of academic ... https://link.springer.com/article/10.1007/s44217-026-01275-6
18. A multimodal dataset for understanding the impact of mobile phones ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12313889/
19. Proctoring exams in the new online learning era - Facebook https://www.facebook.com/groups/higheredlearningcollective/posts/746833585947342/
20. (PDF) Real-Time Detection of Examination Malpractices Using ... https://www.researchgate.net/publication/391727591_Real-Time_Detection_of_Examination_Malpractices_Using_Convolutional_Neural_Networks_and_Video_Surveillance_A_Systematic_Review_with_Meta-Analysis
21. Efficient Online Engagement Analytics Algorithm Toolkit That Can ... https://www.mdpi.com/1999-4893/16/2/86
22. [PDF] Secure Assessment Policy - Jersey College https://www.jerseycollege.edu/application/files/9917/7064/6775/Secure_Assessment_Policy_-_February_2026.pdf
23. Is test proctoring software like Examity an invasion of student privacy? https://www.quora.com/Is-test-proctoring-software-like-Examity-an-invasion-of-student-privacy
24. Students are cheating with AI because you haven't updated ... - Reddit https://www.reddit.com/r/Professors/comments/17jmxja/students_are_cheating_with_ai_because_you_havent/
25. Proceedings | CHI 2021 - ACM https://chi2021.acm.org/proceedings

26. Poster Presentations - 2025 - Otolaryngology–Head and Neck Surgery https://aao-hnsfjournals.onlinelibrary.wiley.com/doi/full/10.1002/ohn.982
27. Deep Learning Models for Detecting Cheating in Online Exams https://www.sciopen.com/article/10.32604/cmc.2025.067359
28. Visualization and quantification of eye tracking data for the ... https://www.sciencedirect.com/science/article/pii/S2405844018348709
29. Using eye movements, electrodermal activities, and heart rates to ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12457608/
30. Robot's Social Gaze Affects Conflict Resolution but not Conflict ... https://journalofcognition.org/articles/10.5334/joc.189
31. Enhancing student memorisation through teacher webcam usage ... https://link.springer.com/article/10.1186/s41239-025-00554-w
32. Current concepts in acute vestibular syndrome and video-oculography https://journals.lww.com/co-neurology/fulltext/2022/02000/current_concepts_in_acute_vestibular_syndrome_and.13.aspx
33. Visual Intention Through Gaze Tracking Using Low-Cost Webcam https://ieeexplore.ieee.org/iel7/6287639/9668973/09813721.pdf
34. [PDF] AI in Higher Education Summit Paris - 17 - 18 March 2026 https://papers.ssrn.com/sol3/Delivery.cfm/6441542.pdf?abstractid=6441542&mirid=1
35. A fully automated online EEG wavelet-based learning adaptive ... https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0313076
36. [PDF] Human Factors Design Guidance for Driver-Vehicle Interfaces https://www.nhtsa.gov/sites/nhtsa.gov/files/documents/812360_humanfactorsdesignguidance.pdf
37. Real-Time Exam Analytics: 11 Key Insights | Proctor360 https://proctor360.com/blog/real-time-exam-analytics-critical-insights
38. A Hybrid Deep Learning Approach for Real-Time Cheating ... https://www.sciencedirect.com/org/science/article/pii/S1546221826000743
39. Student Cheating Detection in Higher Education by Implementing ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10142698/
40. Can AI Detect Suspicious Typing Patterns in Real Time? - HackerRank https://www.hackerrank.com/writing/ai-detect-suspicious-typing-patterns-real-time
41. [PDF] Design and Implementation of an E-Exam Cheating Control System ... https://edinburgjournals.org/journals/index.php/journal-of-information-technolog/article/download/610/651/2377
42. (PDF) Automated Detection of Suspicious Behavior During Online ... https://www.researchgate.net/publication/

395715157_Automated_Detection_of_Suspicious_Behavior_During_Online_Exams_Using_Artificial_Intelligence_and_Computer_Vision

43. False Positives: The Hidden Cost Center in Production AI - Medium https://medium.com/@adnanmasood/false-positives-the-hidden-cost-center-in-production-ai-790afc8c1632
44. [PDF] Intelligent Offline Exam Monitoring System for Identifying Suspicious ... https://ijcaonline.org/archives/volume187/number15/holi-2025-ijca-925125.pdf
45. A Survey on Acoustic Side Channel Attacks on Keyboards - ar5iv https://ar5iv.labs.arxiv.org/html/2309.11012
46. Improving Acoustic Side-Channel Attacks on Keyboards Using ... https://arxiv.org/html/2502.09782v1
47. ISO 30107-3 Presentation Attack Detection Test Methodology ... - iBeta https://www.ibeta.com/iso-30107-3-presentation-attack-detection-confirmation-letters/
48. [PDF] Face Liveness Detection Competition (LivDet-Face) - 2024 https://par.nsf.gov/servlets/purl/10577571
49. Biometric Confirmation Letters certificate biometric testing report https://bixelab.com/confirmation-letter/
50. Presentation Attack Detection using iris periocular visual spectrum ... https://www.frontiersin.org/journals/imaging/articles/10.3389/fimag.2024.1478783/full
51. Shufti wins iBeta Level 3 Certification for Passive Liveness Detection https://shuftipro.com/press-release/shufti-wins-ibeta-level-3-certification-for-passive-liveness-detection/
52. IBeta Level 2: Facia Achieved ISO 30107-PAD Compliance https://facia.ai/news/facia-ibeta-level-2-compliant-with-iso-30107-3-presentation-attack-detection-protocols/
53. iBeta Certification Requirements (Level 1, 2, 3) - Axon Labs https://axonlab.ai/ibeta-certification-requirements-overview/
54. What is iBeta Biometric Testing? A Complete Guide to Face ... https://iapp.co.th/blog/what-is-ibeta-biometric-testing-complete-guide
55. iBeta Biometrics Testing: Everything about the Test - Surfing.ai https://www.surfing.ai/ibeta-biometrics-testing/
56. Introduction to Liveness Detection for Digital Identity - Paravision AI https://www.paravision.ai/news/introduction-to-liveness-detection-for-digital-identity/
57. Top 15 Remote Proctoring Tools - Sherlock AI https://www.withsherlock.ai/blog/remote-proctoring-tools
58. Interview Proctoring for Secure Interviews Lights. Camera. No ... https://www.instagram.com/reel/DLErC06TAtm/

59. How AI is changing the job interview game | Sanjoe Tom Jose ... https://www.linkedin.com/posts/sanjoetomjose_ai-has-quietly-changed-the-rules-of-interviewing-activity-7386465000137162753-2RT-
60. Why Is Virtual Interview Fraud Spiking, and How Do You Stop It? https://recruitmentsmart.com/blogs/why-is-virtual-interview-fraud-spiking-and-how-do-you-stop-it
61. Secure assessment: Innovations in online exam proctoring - Turnitin https://www.turnitin.com/blog/secure-assessment-innovations-in-online-exam-proctoring
62. How does MediaPipe achieve fast inference speed? - Facebook https://www.facebook.com/groups/ComputerVisionGroup/posts/2454989254645261/
63. AI Inference Pipelines – Building Low-Latency Systems With gRPC https://www.youtube.com/watch?v=ISLGPZ493MI
64. [PDF] Enhancing Real-time Inference Performance for Time-Critical ... https://sidilu.org/paper/Enhancing_Real-time_Inference_Performance_for_Time-Critical_Software-Defined_Vehicles.pdf
65. Revolutionizing Real-Time AI with NVIDIA TensorRT, DRP-AI TVM ... https://www.tmasolutions.com/insights/ai-on-edge-device-revolutionizing-real-time-ai-with-nvidia-tensorrt-drp-ai-tvm-and-snpe
66. Ai-based Online Interview Behavioral Analysis - IJSART https://ijsart.com/ai-based-online-interview-behavioral-analysis-an-ai-driven-approach-to-proctoring-and-integrity-verification-105210
67. Talview Interview Proctoring - Microsoft Marketplace https://marketplace.microsoft.com/en-us/product/saas/talview.talview_interview_proctoring?tab=overview
68. Leveraging Multimodal Behavioral Analytics for Automated Job ... https://aclanthology.org/2020.challengehml-1.6/
69. Talview: World's First Agentic AI for Proctoring and Interviewing https://www.talview.com/en/
70. What is Audio Proctoring? - iMocha https://www.imocha.io/glossary/what-is-audio-proctoring
71. Leveraging Multimodal Behavioral Analytics for Automated Job ... https://arxiv.org/abs/2006.07909
72. Top 15 Interview Proctoring Tools in 2026 - Sherlock AI https://www.withsherlock.ai/blog/interview-proctoring-tools
73. Robust audio anomaly detection - Amazon Science https://www.amazon.science/publications/robust-audio-anomaly-detection
74. Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable ... https://arxiv.org/html/2504.11622v1

75. A Survey on Acoustic Side Channel Attacks on Keyboards https://www.researchgate.net/publication/374167346_A_Survey_on_Acoustic_Side_Channel_Attacks_on_Keyboards
76. [PDF] Stealing Keystroke Content from Keyboard Acoustic Side-Channel https://papers.ssrn.com/sol3/Delivery.cfm/95c01cdf-40f7-4fe1-a8e5-5a9408927b70-MECA.pdf?abstractid=5943496&mirid=1
77. SKAID: A Realistic Dataset for Acoustic Side-Channel Attacks with ... https://zenodo.org/records/17282184
78. A Survey on Acoustic Side-Channel Attacks: An Artificial Intelligence ... https://www.mdpi.com/2624-800X/6/1/6
79. A Practical Deep Learning-Based Acoustic Side Channel Attack on ... https://github.com/ggerganov/kbd-audio/discussions/58
80. Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable ... https://www.usenix.org/conference/woot25/presentation/ayati
81. Introducing Whisper - OpenAI https://openai.com/index/whisper/
82. collabora/WhisperLive: A nearly-live implementation of OpenAI's ... https://github.com/collabora/WhisperLive
83. WhisperX: Time-Accurate Speech Transcription of Long-Form Audio https://arxiv.org/html/2303.00747v2
84. Top APIs and models for real-time speech recognition ... - AssemblyAI https://www.assemblyai.com/blog/best-api-models-for-real-time-speech-recognition-and-transcription
85. Whisper AI - Professional Voice to Text Transcription https://whisperai.com/
86. Best STT Providers 2026: Independent Benchmarks & How to Choose https://www.coval.ai/blog/best-speech-to-text-providers-in-2026-independent-benchmarks-and-how-to-choose/
87. Whisper Speech-to-Text Alternative for Real-time Transcription https://picovoice.ai/blog/whisper-speech-to-text-for-real-time-transcription/
88. Whisper, A Breakthrough in Speech-Recognition AI - ENERZAi https://enerzai.com/resources/blog/whisper-a-breakthrough-in-speech-recognition-ai
89. Automatic speech recognition and the transcription of indistinct ... https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2024.1281407/full
90. Infrared Detection of Elevations in Mobile Phone Temperatures ... https://www.researchgate.net/publication/338443022_Infrared_Detection_of_Elevations_in_Mobile_Phone_Temperatures_Induced_by_Casings

91. Infrared Detection of Elevations in Mobile Phone Temperatures ... https://www.academia.edu/51930246/Infrared_Detection_of_Elevations_in_Mobile_Phone_Temperatures_induced_by_Casings
92. [PDF] Revealing Hidden IoT Devices through Passive Detection ... https://petsymposium.org/popets/2025/popets-2025-0011.pdf
93. Revealing hidden IoT devices through passive detection ... - SOAR https://soar.wichita.edu/entities/publication/4847b779-63f9-47d0-a6ef-6b62e63fdd10
94. Enhancing Wireless Device Identification through RF Fingerprinting https://www.researchgate.net/publication/392941347_Enhancing_Wireless_Device_Identification_through_RF_Fingerprinting_Leveraging_Transient_Energy_Spectrum_Analysis
95. [PDF] AI-BASED RF FINGERPRINTING DEVICE IDENTIFICATION AND ... https://ajird.journalspark.org/index.php/ajird/article/download/880/846/874
96. (PDF) Multi-sensor fusion for user matching and continuous tracking ... https://www.researchgate.net/publication/404685132_Multi-sensor_fusion_for_user_matching_and_continuous_tracking_leveraging_camera_handover_in_surveillance_systems
97. ECCV Conference Papers - ECVA https://www.ecva.net/papers.php
98. Multi-sensor fusion and segmentation for autonomous vehicle multi ... https://www.nature.com/articles/s41598-024-82356-0
99. A Practical Deep Learning-Based Acoustic Side Channel Attack on ... https://www.researchgate.net/publication/372799628_A_Practical_Deep_Learning-Based_Acoustic_Side_Channel_Attack_on_Keyboards
100. Improved CoAtNet for robust acoustic side-channel attack ... https://link.springer.com/article/10.1007/s10207-025-01194-x
101. Privacy-Preserving Machine Learning Techniques: Cryptographic ... https://www.mdpi.com/2076-3417/16/1/277
102. Approximate homomorphic encryption based privacy-preserving ... https://link.springer.com/article/10.1007/s10462-024-11076-8
103. Architecting Neural Networks for Fast Secure Inference with ... - arXiv https://arxiv.org/html/2601.21287v1
104. Homomorphic encryption protocols and applications in privacy ... http://ictp.caict.ac.cn/EN/abstract/abstract934.shtml
105. ISO 27001 AI Compliance – Information Security for AI Systems https://aihealthcarecompliance.com/resources/applicable-laws/iso-27001/
106. Recruitment Smart Trust Center https://recruitmentsmart.com/security-compliance-ai-ethics

107. What Is ISO 27001 and Why Does It Matter for AI Vision? - Viso Suite https://viso.ai/product/iso-27001-computer-vision/

108. [PDF] Security Standards Compliance NIST SP 800-53 Revision 5 ... https://www.dlt.com/sites/default/files/resource-attachments/2019-09/Whitepaper%2520SP800-53r5-TM_Compliancy_v3.2_9May18_0_13.pdf

109. AU-11: Audit Record Retention - CSF Tools https://csf.tools/reference/nist-sp-800-53/r5/au/au-11/

110. AI can now take online exams, mimic human behavior ... - Instagram https://www.instagram.com/reel/DYmctoEjmdu/

111. talview - Instagram https://www.instagram.com/p/DExBH6LyQNc/

112. [PDF] Lips Are Lying: Spotting the Temporal Inconsistency between Audio ... https://proceedings.neurips.cc/paper_files/paper/2024/file/a5a5b0ff87c59172a13342d428b1e033-Paper-Conference.pdf

113. [PDF] Lip-Sync Deepfake Detection from Audio-Video Mismatch https://openaccess.thecvf.com/content/CVPR2024W/WMF/papers/Bohacek_Lost_in_Translation_Lip-Sync_Deepfake_Detection_from_Audio-Video_Mismatch_CVPRW_2024_paper.pdf

114. Lips Are Lying: Spotting the Temporal Inconsistency between Audio... https://openreview.net/forum?id=yMS7ansbr6&referrer=%5Bthe%20profile%20of%20Boheng%20Li%5D(%2Fprofile%3Fid%3D~Boheng_Li1)

115. skrantidatta/LIPINC - GitHub https://github.com/skrantidatta/LIPINC

116. Detecting Lip-Syncing Deepfakes: Vision Temporal Transformer for ... http://ui.adsabs.harvard.edu/abs/2025arXiv250401470K/abstract

117. [PDF] Future-Proofing Deepfake Detection by Integrating Audio, Video ... https://www.eurecom.fr/publication/8672/download/data-publi-8672.pdf

118. WiPIHT: A WiFi-Based Position-Independent Passive Indoor Human ... https://www.mdpi.com/1424-8220/25/13/3936

119. Bluetooth Scanner in Proctoring: How to Detect Hidden Cheating ... https://trustexam.ai/blog/bluetooth-scanner-in-proctoring-how-to-detect-hidden-cheating-devices

120. AI Proctoring Detect Phones in Online Exams & Prevent Mobile ... https://www.talview.com/blog/ai-proctoring-detects-cellphones

121. Revealing Hidden IoT Devices through Passive Detection ... https://petsymposium.org/popets/2025/popets-2025-0011.php

122. [PDF] Real-Time Depth Estimation Using On-Device Dual Cameras https://www.microsoft.com/en-us/research/wp-content/uploads/2022/09/mobicom22-final138.pdf

123. [PDF] Integrated Multi-Stereo Camera System for Robust Indoor ... - SerWisS https://serwiss.bib.hs-hannover.de/files/3454/mortazavi_etal2024-multi-stereo_camera_system.pdf

124. [PDF] Whisper+AASIST for DeepFake Audio Detection https://faculty.cs.gwu.edu/xiaodongqu/papers/HCII_2024_572_Whisper_Qian.pdf

125. What Is P99 Latency? Understanding the 99th Percentile of ... https://aerospike.com/blog/what-is-p99-latency/

126. P99 Latency: What It Means & How to Fix It - Redis https://redis.io/blog/p99-latency/

127. What AI Proctoring Means for Secure Certification Exams https://www.taotesting.com/blog/ai-proctoring/

128. Privacy-preserving Approaches for Biometric Recognition Systems https://anr.fr/Project-ANR-20-CE39-0005

129. Top AI Proctoring Platforms with GDPR Compliance in 2026 - Proview https://proview.io/blog/top-ai-proctoring-platforms-with-gdpr-compliance-in-2026

130. Regulatory Compliance details for NIST SP 800-53 Rev. 5 https://learn.microsoft.com/en-us/azure/governance/policy/samples/nist-sp-800-53-r5

131. Deepfake Detection Methods 2026 - UncovAI https://uncovai.com/deepfake-detection-methods-2026/

132. [PDF] Lip-Sync Deepfake Detection from Audio-Video Mismatch - Hany Farid https://farid.berkeley.edu/downloads/publications/cvprw24c.pdf

133. SynthGuard - CVPR 2026 Open Access Repository https://openaccess.thecvf.com/content/CVPR2026W/AIMS/html/Desai_SynthGuard_An_Open_Platform_for_Detecting_AI-Generated_Multimedia_with_Multimodal_CVPRW_2026_paper.html

134. [PDF] Advanced AI Examination Monitoring and Security System https://www.scitepress.org/Papers/2025/136127/136127.pdf

135. [PDF] AI-BASED ONLINE EXAM PROCTORING SYSTEM USING MULTI ... https://ijcrt.org/papers/IJCRT2604663.pdf

136. Face 2024 - LivDet https://face2024.livdet.org/

137. (PDF) Face Liveness Detection Competition (LivDet-Face) - 2024 https://www.researchgate.net/publication/385727961_Face_Liveness_Detection_Competition_LivDet-Face_-_2024

138. Press Release: Neurotechnology Achieves Compliance for ... https://www.neurotechnology.com/press_release_pad_level_2_compliance.html

139. ISO/IEC DIS 30107-3(en), Information technology https://www.iso.org/obp/ui/#iso:std:iso-iec:30107:-3:dis:ed-1:v1:en:sec:6

140. iBeta Level 3 and the New Standards - Unidata https://unidata.pro/blog/ibeta-level-3-new-standards/

141. [PDF] Presentation Attack Detection - ISO/IEC 30107 - Christoph Busch https://www.christoph-busch.de/files/Busch-PAD-240701.pdf

142. [PDF] Face Liveness Detection Competition (LivDet-Face) - 2024 https://www.semanticscholar.org/paper/Face-Liveness-Detection-Competition-(LivDet-Face)-Igene-Hossain/34ef0bb47e09678451c2b8b767b936c8072a6f19

143. Liveness detection metrics — Antispoofing Wiki https://antispoofing.org/antispoofing-performance-metrics-types-and-details/

144. Exam Proctoring Software with Webcam in 2026 https://thinkexam.com/blog/top-exam-proctoring-software-with-webcam-2026-how-ai-driven-webcam-monitoring-prevents-cheating/

145. How We Built a Real-Time AI Proctoring System - Medium https://medium.com/@dhimanshashank15/how-we-built-a-real-time-ai-proctoring-system-and-watched-it-collapse-at-five-users-d8e14202e8d1

146. Program - AI in Clinical trials Symposium - Sciencesconf https://ai-trials-2026.sciencesconf.org/page/program?lang=en

147. AI-based online proctoring: a review of the state-of-the-art ... https://www.researchgate.net/publication/374004866_RETRACTED_ARTICLE_AI-based_online_proctoring_a_review_of_the_state-of-the-art_techniques_and_open_challenges

148. Can AI Handle Exam Proctoring Without Human Watching? - LinkedIn https://www.linkedin.com/posts/proctortrack-pro_aiproctoring-automatedproctoring-remoteassessments-activity-7357085348666986496-dWjN

149. ASPAI 2026: 8th International Conference on Advances in Signal ... https://www.aspai-conference.com/

150. AAAI.2026 | Cool Papers - Immersive Paper Discovery https://papers.cool/venue/AAAI.2026

151. Competitions - IJCAI 2026 https://2026.ijcai.org/competitions/

152. Optimizing Low-Latency Video Streaming - ACM SIGCOMM 2025 https://conferences.sigcomm.org/sigcomm/2025/tutorials-hackathons/tutorial-vsai

153. A Detection and Recognition Method for Interference Signals Based ... https://www.mdpi.com/2079-9292/15/7/1393

154. Infrastructure and Methods for WiFi-Based Passive Device ... https://ui.adsabs.harvard.edu/abs/2024PhDT.....64M/abstract

155. Online Proctored Dual Camera Requirement - Kryterion Support https://support.kryterion.com/en_us/online-proctored-dual-camera-requirement-rJJnB8Jo1l

156. 360° Total View Proctoring: Tech Guide | Proctor360 https://proctor360.com/blog/360-degree-online-proctoring-total-view-technology-guide
157. Remote Proctoring with AI: The Future of Online Exams - HackerEarth https://www.hackerearth.com/blog/remote-proctoring-online-assessments
158. ISO 27001:2022 Annex A 8.16 – Monitoring Activities - ISMS.online https://www.isms.online/iso-27001/annex-a-2022/8-16-monitoring-activities-2022/
159. NIST SP 800-12: Chapter 18 - Audit Trails - CSRC https://csrc.nist.rip/publications/nistpubs/800-12/800-12-html/chapter18.html
160. ISO 27001:2022 Annex A 8.15 – Logging - ISMS.online https://www.isms.online/iso-27001/annex-a-2022/8-15-logging-2022/
161. [PDF] Guide to Computer Security Log Management https://nvlpubs.nist.gov/nistpubs/legacy/SP/nistspecialpublication800-92.Pdf
162. TRACE - CVPR 2026 Open Access Repository https://openaccess.thecvf.com/content/CVPR2026W/MMFM5/html/khan_TRACE_Training-Free_Partial_Audio_Deepfake_Detection_via_Embedding_Trajectory_Analysis_CVPRW_2026_paper.html
163. [PDF] Privacy Preserving Gaze Estimation using Synthetic Images via a ... https://arxiv.org/pdf/1911.07936
164. [PDF] PrivateEyes: Gaze-Preserving Anonymization for Data Sharing https://openaccess.thecvf.com/content/CVPR2026/papers/Gupta_PrivateEyes_Gaze-Preserving_Anonymization_for_Data_Sharing_CVPR_2026_paper.pdf
165. Privacy-aware eye tracking using differential privacy https://www.semanticscholar.org/paper/Privacy-aware-eye-tracking-using-differential-Steil-Hagestedt/97042ab431354860985e708bc6551ad3dfe5e3e4
166. [PDF] A privacy-preserving approach to streaming eye-tracking data https://par.nsf.gov/servlets/purl/10354953
167. Differential privacy for eye tracking with temporal correlations - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8370645/
168. Privacy-aware eye tracking using differential privacy - ResearchGate https://www.researchgate.net/publication/333490966_Privacy-aware_eye_tracking_using_differential_privacy
169. Thermal Anomaly Area Intelligent Recognition Method Based on ... https://www.authorea.com/doi/10.22541/au.169028989.93419503
170. [PDF] Detection of Subtle Thermal Anomalies: Deep Learning Applied to ... https://uca.hal.science/hal-04802870v1/file/Detection_of_Subtle_Thermal_Anomalies_Deep_Learning_Applied_to_the_ASTER_Global_Volcano_Dataset.pdf

171. Complete Guide To Thermal Anomaly Detection (2026) - Averroes AI https://averroes.ai/blog/thermal-anomaly-detection-guide
172. Real-Time Video Super-Resolution - CodaLab - Competition https://codalab.lisn.upsaclay.fr/competitions/21558
173. [PDF] AI-BASED SMART EXAMINATION PROCTORING SYSTEM - IRJMETS https://www.irjmets.com/upload_newfiles/irjmets80400106831/paper_file/irjmets80400106831.pdf
174. [PDF] 2025 Face Liveness Market Report and Buyer's Guide https://www.biometricupdate.com/wp-content/uploads/2025/11/Biometric-Update-2025-Facial-Liveness-Report.pdf
175. What is iBeta certification? - Unidata https://unidata.pro/blog/ibeta-certification/
176. Mobile Contactless Fingerprint Presentation Attack Detection https://ieeexplore.ieee.org/iel7/8423754/10767127/10536028.pdf
177. [PDF] DyFFPAD: Dynamic Fusion of Convolutional and Handcrafted ... https://arxiv.org/pdf/2308.10015
178. Stereo Depth Cameras - RealSense https://www.realsenseai.com/stereo-depth-cameras/
179. Drools rule engine https://docs.drools.org/latest/drools-docs/drools/rule-engine/index.html
180. Everything Old Is New Again: Rule Engines in the Age of Agents https://medium.com/@adnanmasood/everything-old-is-new-again-rule-engines-in-the-age-of-agents-9a20c9459556
181. Top 10 Open Source Rule Engines in 2026: What Actually Works https://levelup.gitconnected.com/top-10-open-source-rule-engines-in-2026-what-actually-works-2b6a59a64366
182. Chapter 6. Complex event processing (CEP) - Red Hat Documentation https://docs.redhat.com/en/documentation/red_hat_decision_manager/7.6/html/decision_engine_in_red_hat_decision_manager/cep-con_decision-engine
183. Acoustic Side-Channel Vulnerabilities in Keyboard Input Explored ... https://www.mdpi.com/2076-3417/16/2/563
184. SoK: Acoustic Side Channels | ACM Computing Surveys https://dl.acm.org/doi/10.1145/3778350
185. Synthetic Voice Detection and Audio Splicing Detection using SE ... https://arxiv.org/abs/2210.03581
186. AI Proctoring & Fraud Detection for Online Interviews | Intrvuz https://www.intrvuz.com/ai-proctoring

187. Synthetic Voice Detection and Audio Splicing Detection using SE ... https://www.semanticscholar.org/paper/Synthetic-Voice-Detection-and-Audio-Splicing-using-Wang-Yeoh/561b699c1f25d77b50f299241ec045083351f2fa

188. A Comprehensive Survey on Radio Frequency (RF) Fingerprinting https://www.techrxiv.org/doi/10.36227/techrxiv.17711444

189. [PDF] Improved DeepFake Detection Using Whisper Features - ISCA Archive https://www.isca-archive.org/interspeech_2023/kawa23b_interspeech.pdf

190. [PDF] vSGX: Virtualizing SGX Enclaves on AMD SEV - Shixuan Zhao's https://www.nskernel.org/assets/files/vSGX-Virtualizing-SGX-Enclaves-on-AMD-SEV.pdf

191. [PDF] Horizontal Federated Learning and Secure Distributed Training for ... https://arxiv.org/pdf/2207.05079

192. IACR News https://iacr.org/news/index.php?next=27640

193. [PDF] special icpR 2024 issue! https://iapr.org/docs/newsletters/2025-Vol-47-No-2.pdf

194. [PDF] Applications received during 1 to 28 February, 2026 - IP India https://ipindia.gov.in/storage/uploads/docs-operator/7ca64652-f23d-4389-b7b7-7c6d11fc65b6.pdf

195. [PDF] Plastic Degrading Enzymes and Microbes https://tnsroindia.org.in/JOURNAL/issue83/IJONS%20-%20ISSUE%2083%20-%20APRIL%202024%20-%20FULL%20TEXT%20%20-%20PART%20-%2002.pdf

196. [PDF] національна академія наук україни https://ipme.kiev.ua/wp-content/uploads/2025/12/%D0%97%D0%B1%D1%96%D1%80%D0%BD%D0%B8%D0%BA-%D0%BC%D0%B0%D1%82%D0%B5%D1%80%D1%96%D0%B0%D0%BB%D1%96%D0%B2-%D0%A8%D0%86%D0%91-2025.pdf

197. [PDF] 14 January 2026 - iBeta https://www.ibeta.com/wp-content/uploads/2026/01/260114-Compal-PAD-Level-2-Confirmation-Letter-Dual-Camera.pdf

198. [PDF] 23 January 2026 - iBeta https://www.ibeta.com/wp-content/uploads/2026/01/260123-Yoti-PAD-Level-3-Confirmation-Letter-1.pdf

199. Forensic deepfake audio detection using segmental speech features https://arxiv.org/html/2505.13847v3

200. [PDF] Deepfake Audio Detection via Feature Engineering and Machine ... https://ceur-ws.org/Vol-3318/paper4.pdf

201. A Practical Deep Learning-Based Acoustic Side Channel Attack on ... https://www.semanticscholar.org/paper/01882baf1fcce4883a60fb22db9a3f6ae9b493ad

202. 3 Key Strategies for Noisy Speech Recognition in 2026 (WER ... https://www.forasoft.com/blog/article/speech-recognition-accuracy-noisy-environments

203. Exploring OpenAI Whisper Speech Recognition - Deepgram Blog ⚡ https://deepgram.com/learn/exploring-whisper
204. Whisper vs Gemini vs AssemblyAI vs Deepgram - Codesota https://www.codesota.com/guides/speech-recognition
205. AU-2 - NIST 800-53 r5 Control Explorer - GRC Academy https://grcacademy.io/nist-800-53/controls/au-2/
206. Anonymization Techniques for Behavioral Biometric Data: A Survey https://www.researchgate.net/publication/390928546_Anonymization_Techniques_for_Behavioral_Biometric_Data_A_Survey
207. PrivateEyes: Gaze-Preserving Anonymization for Data Sharing https://cvpr.thecvf.com/virtual/2026/poster/37919
208. AI Hiring Compliance - GDPR DPDP SOC 2 ISO 27001 - X0PA AI https://x0pa.com/compliance/
209. AU-2: Event Logging - CSF Tools https://csf.tools/reference/nist-sp-800-53/r5/au/au-2/
210. The Challenge on General AIGC Audio–Video Detection - Codabench https://www.codabench.org/competitions/15769/
211. Secure Audit Logging: An overview - Pangea Cloud https://pangea.cloud/securebydesign/secure-audit-logging-overview/
212. 3.3: Audit and Accountability - CSF Tools https://csf.tools/reference/nist-sp-800-171/r2/3-3/
213. Privacy and biometric data protection: Biometrics and GDPR https://www.mobbeel.com/en/blog/privacy-and-biometric-data-protection-biometrics-and-gdpr/
214. NIST SP 800-53 AU-9: Immutable Audit Logs & Protection https://feeds.trussed.ai/blog/nist-sp-800-53-au-9-protection-audit-information-immutable-logs
215. A Secure Stream Analytics Framework In SGX-enabled Edge Cloud https://www.researchgate.net/publication/367763193_SGX-Stream_A_Secure_Stream_Analytics_Framework_In_SGX-enabled_Edge_Cloud